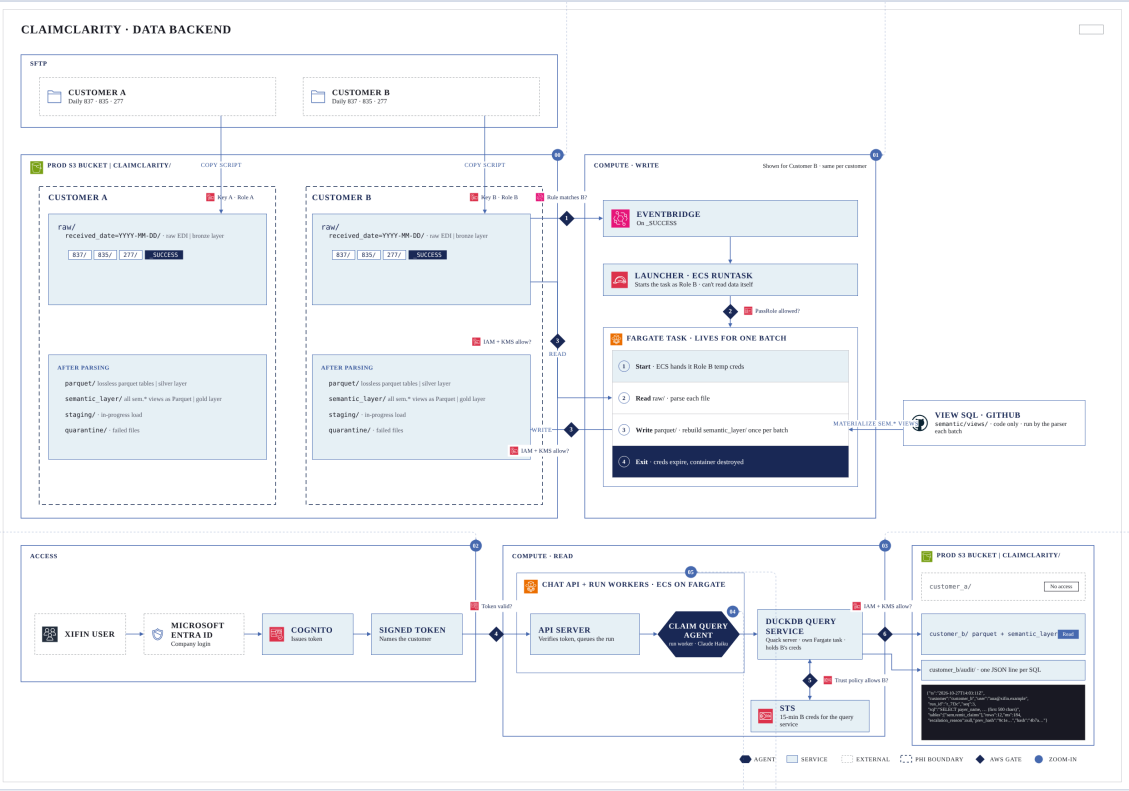
ClaimClarity architecture

ClaimClarity answers questions about a clinical lab's claims (837P) and remittances (835). Each customer's data sits in its own S3 prefix, under its own KMS key and IAM roles. Six AWS checks, the numbered diamonds, enforce that on every write and read. The main board shows the whole system, and drill-downs 00–05 zoom into each part.

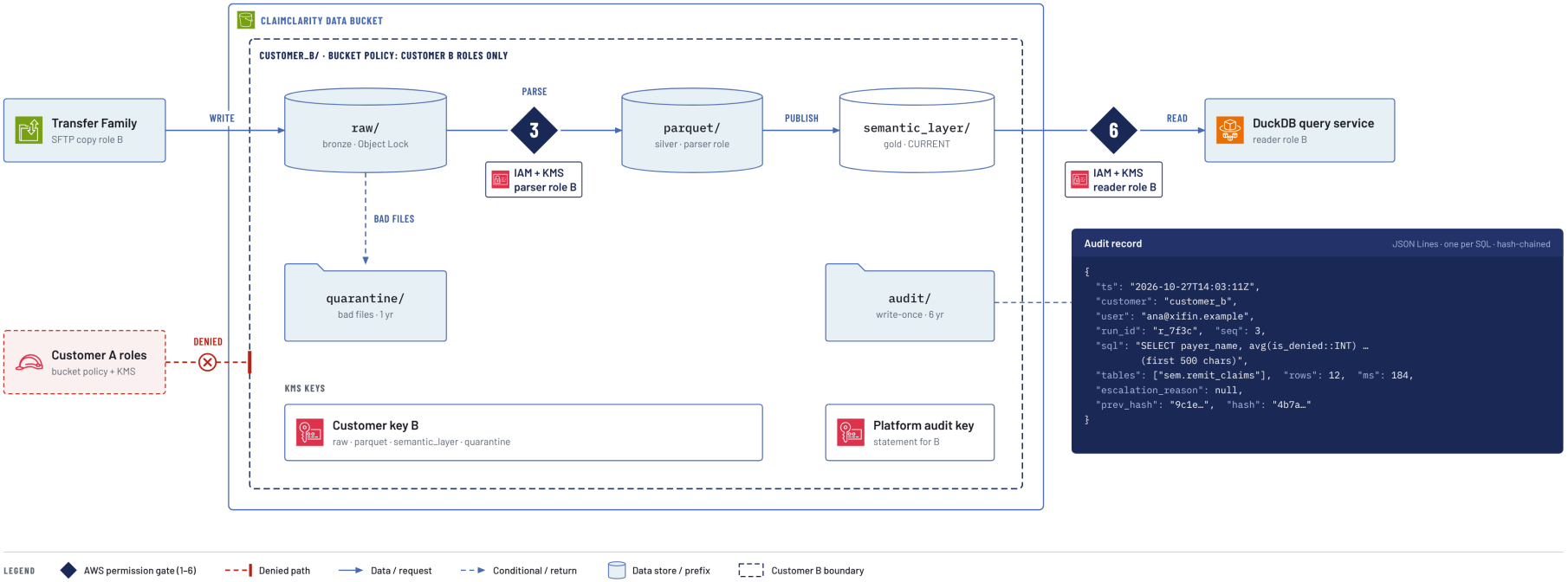
Gate	What AWS checks
1	The EventBridge rule matches the customer
2	`iam:PassRole` allows only the customer's parser role
3	IAM + KMS for the parser
4	The token is valid
5	The reader role's trust policy allows this token (STS)
6	IAM + KMS for the reader

Data backend



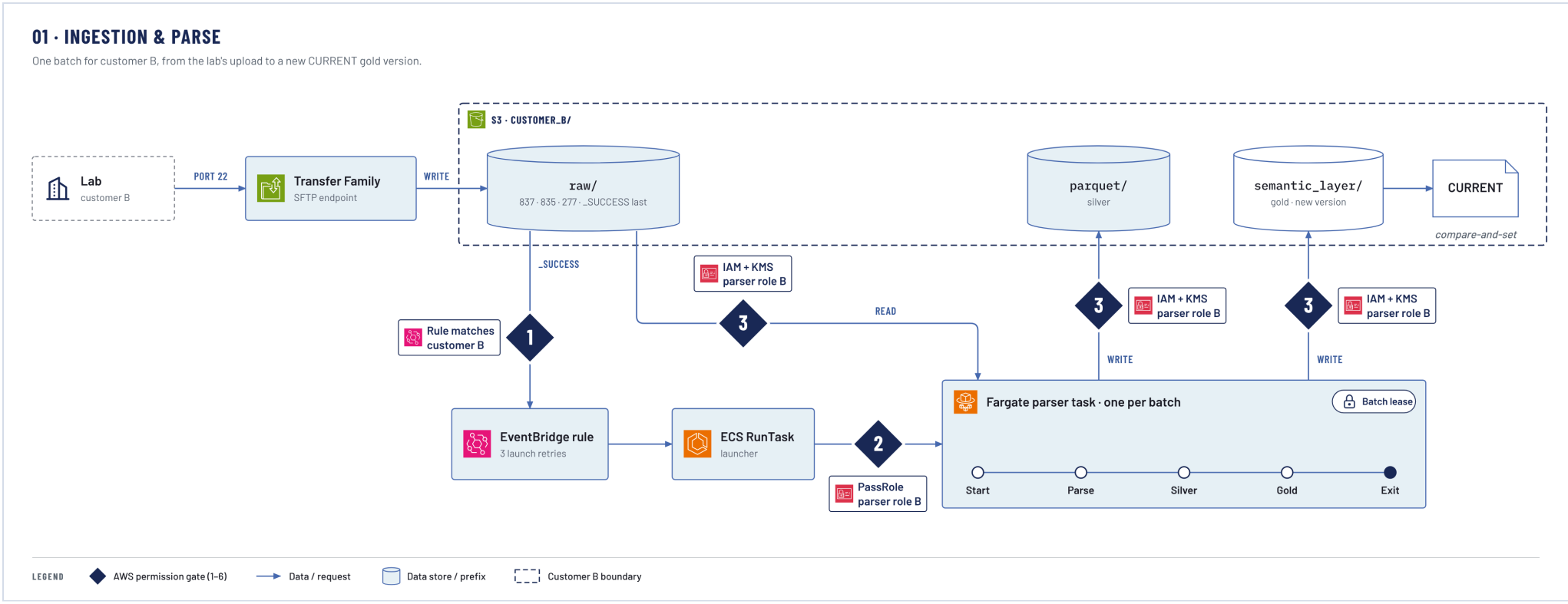
Part	What it does	Key facts
SFTP	Each lab sends daily 837 · 835 · 277 files	AWS Transfer Family, lab IP addresses only, straight into `raw/`
ClaimClarity data bucket	One prefix per customer: raw → silver → gold, plus quarantine and audit	One bucket for all customer data
Compute · write	EventBridge → ECS RunTask → one Fargate parser task per batch	Runs as the customer's parser role. Gates 1–3.
Access	Entra ID → Cognito → ID token naming the customer	The token is the only source of the customer. Gate 4.
Compute · read	Chat API → run worker (the Claim Query Agent) → DuckDB query service → STS → gold	15-minute credentials, this customer's gold only. Gates 5–6.

00 · Storage & isolation



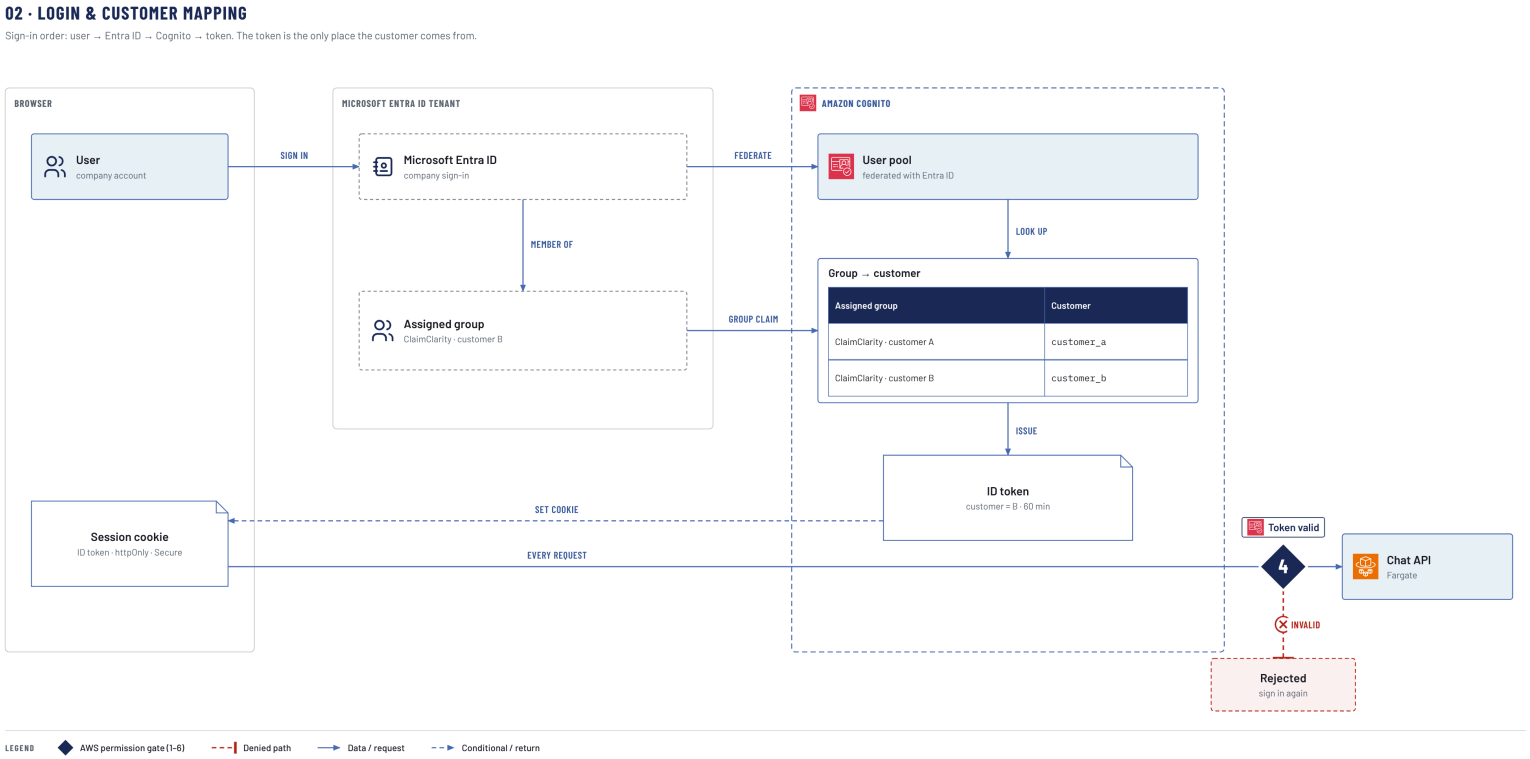
Part	Key facts
Bucket	One bucket for all customer data, one prefix per customer. The web app and the access logs have their own buckets.
Prefixes	`raw/` (write-once, Object Lock 7 years, `_SUCCESS` written last), `parquet/` (silver), `semantic_layer/` (gold), `quarantine/` (bad files, kept 1 year), `audit/`
IAM roles	Per customer: SFTP copy, parser, reader, audit. IAM prefix policies keep the roles apart.
KMS keys	The customer key, one per customer, covers raw, silver, gold and quarantine. The platform audit key covers `audit/`, with one key-policy statement per customer.
Bucket policy	Only customer B's roles reach `customer_b/`. Customer A's roles are denied by the bucket policy and the key policy.
Query audit	The DuckDB query service writes one JSON Lines record per SQL statement: write-once, kept 6 years, each record chained by hash to the one before
Gates	3 on the parser's reads and writes, 6 on the reader's reads

01 · Ingestion & parse



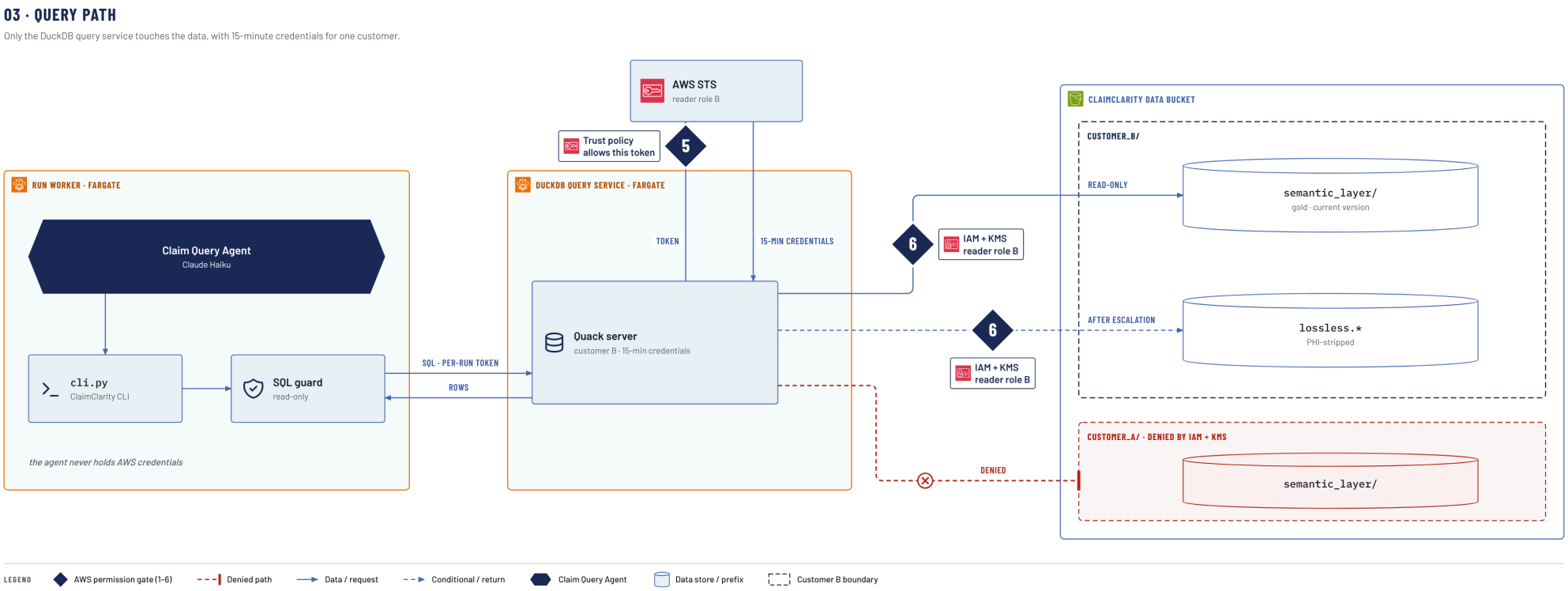
Step	What happens
Intake	The lab uploads over SFTP (port 22, lab IP addresses only). Transfer Family writes straight into `raw/` as the SFTP copy role, _SUCCESS last.
Gate 1	The customer's EventBridge rule matches _SUCCESS under `customer_b/raw/` only
Launch	ECS RunTask starts one parser task. EventBridge retries the launch 3 times.
Gate 2	`iam:PassRole` lets the launcher pass only customer B's parser role, and only to ECS tasks
Parser task	Fargate, 4–8 GB of memory, one per batch. It holds the batch lease, a lock object, while it runs. Bad files go to `quarantine/`.
Gate 3	IAM + KMS on every read of `raw/` and every write to silver and gold
Publish	Gold is written as a new version, and `CURRENT` moves by compare-and-set. Old versions stay readable for sessions already open.

02 · Login & customer mapping



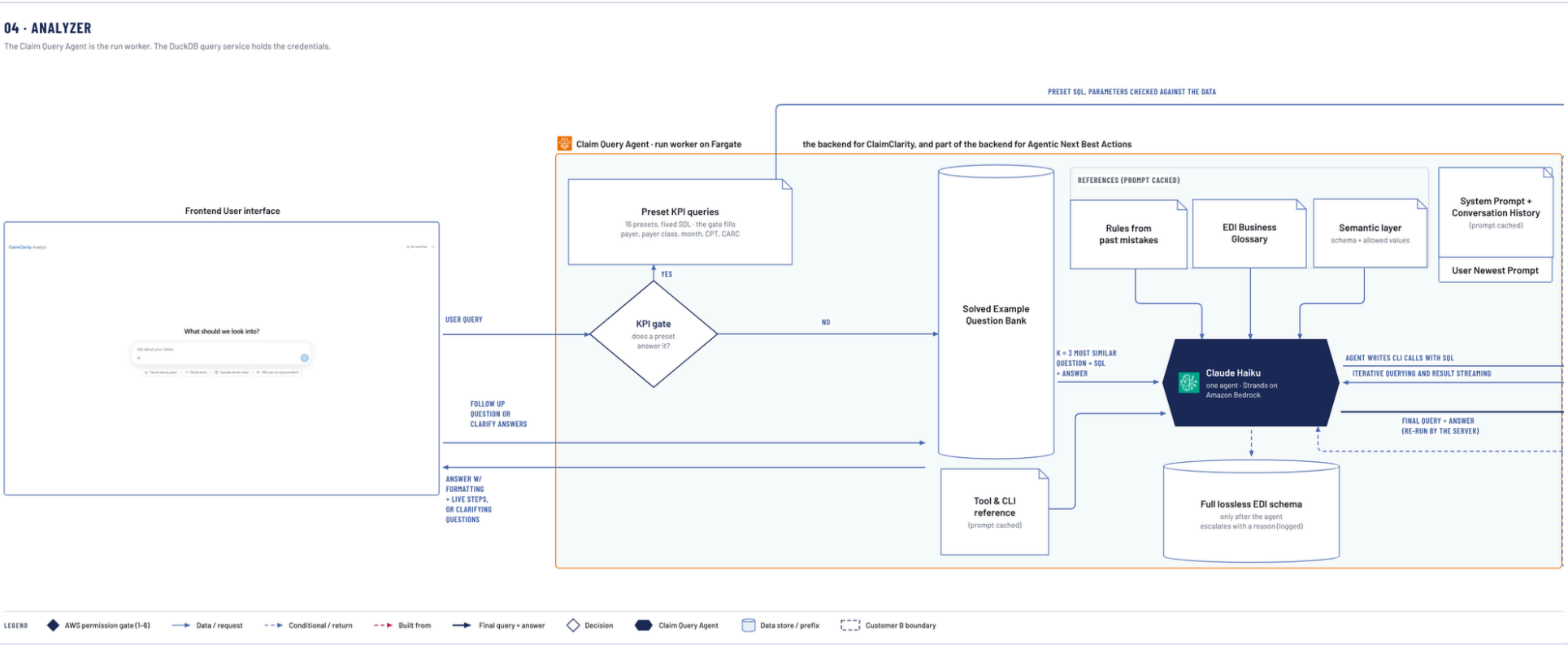
Part	Key facts
Microsoft Entra ID	Company sign-in. The ClaimClarity app is assigned to groups.
Assigned group	One group per customer
Amazon Cognito	Federates Entra ID, maps the group to exactly one customer, and issues an ID token naming that customer, valid 60 minutes
Session cookie	httpOnly and Secure. It carries the ID token.
Gate 4	The chat API checks the token's signature, audience, expiry and customer on every request. An invalid token is rejected.

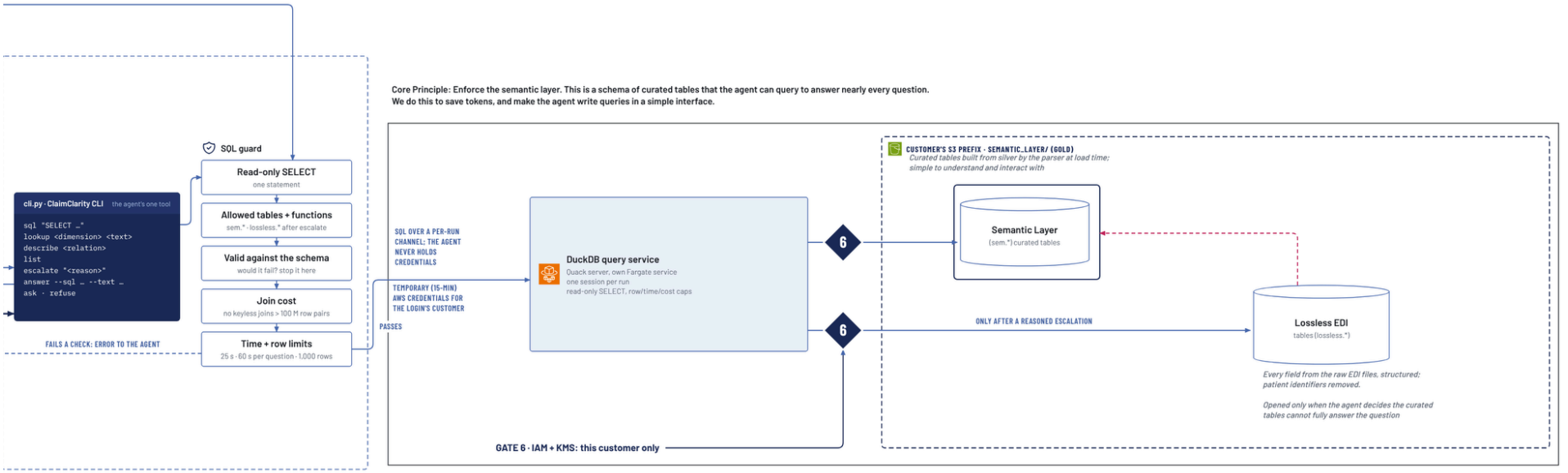
03 · Query path



Part	Key facts
Run worker	The Claim Query Agent sends SQL through `cli.py` and the SQL guard. It holds no AWS credentials.
SQL guard	Read-only SELECT. Blocks queries that would fail against the schema. 25 s per query, 60 s per question, 1,000 rows. Refuses keyless joins over 100 million row pairs.
Quack	SQL goes to the DuckDB query service over Quack with a per-run token, and rows come back the same way
Gate 5	The reader role's trust policy accepts only this token's customer. STS returns read-only credentials for 15 minutes, renewed with the same token when fewer than 3 minutes remain.
Gate 6	IAM + KMS allow reader role B only `customer_b/semantic_layer/`, current gold version
`lossless.*`	PHI-stripped tables published next to gold, read only after the agent escalates with a reason
Customer A	Denied twice: the API takes the customer only from the token, and IAM + KMS deny reader role B on `customer_a/`

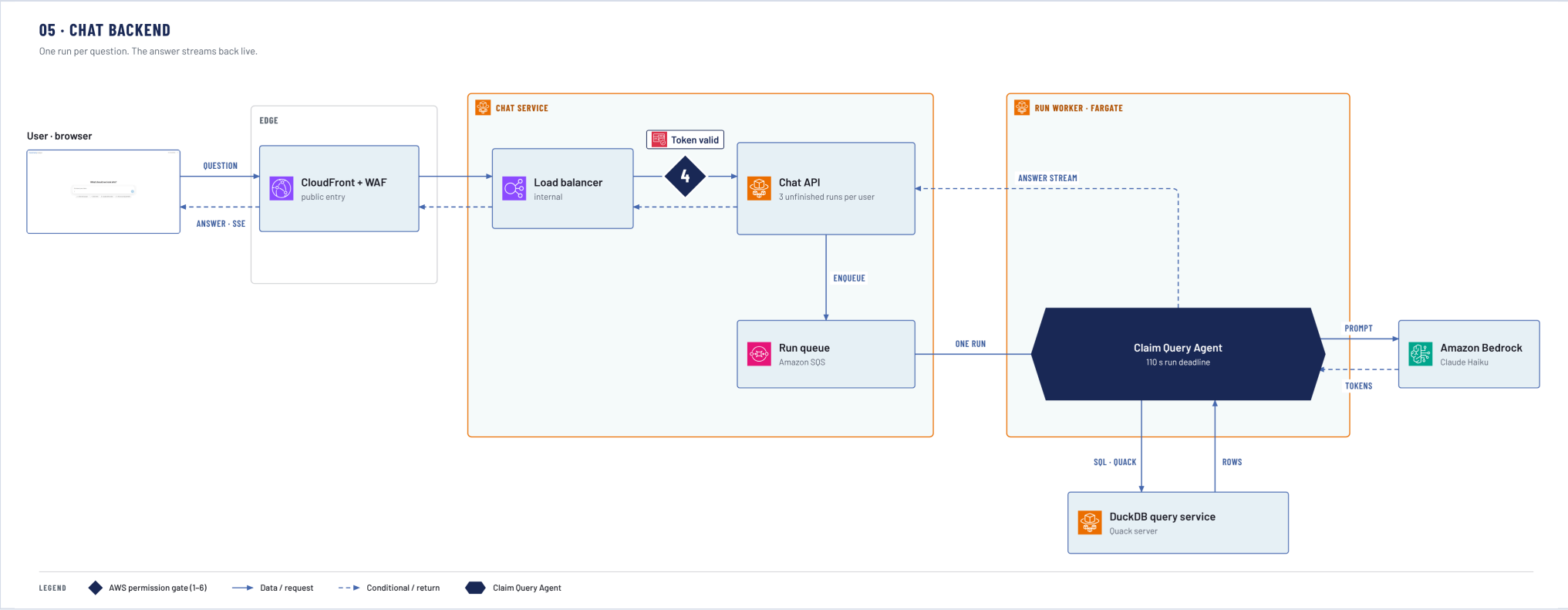
04 · Analyzer


04 · ANALYZER
The Claim Query Agent is the run worker. The DuckDB query service holds the credentials.
PRESET SQL, PARAMETERS CHECKED AGAINST THE DATA
Claim Query Agent · run worker on Fargate
the backend for ClaimClarity, and part of the backend for Agentic Next Best Actions
Frontend User interface
What should we look into?
USER QUERY
FOLLOW UP QUESTION OR CLARIFY ANSWERS
ANSWER W/ FORMATTING + LIVE STEPS, OR CLARIFYING QUESTIONS
Preset KPI queries
16 presets, fixed SQL - the gate fills payer, payer class, month, CPT, CARC
YES
KPI gate
does a preset answer it?
NO
Solved Example Question Bank
Tool & CLI reference
(prompt cached)
REFERENCES (PROMPT CACHED)
Rules from past mistakes
EDI Business Glossary
Semantic layer
schema + allowed values
System Prompt + Conversation History
(prompt cached)
User Newest Prompt
K = 3 MOST SIMILAR QUESTION + SQL + ANSWER
Claude Haiku
one agent · Strands on Amazon Bedrock
AGENT WRITES CLI CALLS WITH SQL
ITERATIVE QUERYING AND RESULT STREAMING
FINAL QUERY + ANSWER (RE-RUN BY THE SERVER)
Full lossless EDI schema
only after the agent escalates with a reason (logged)
LEGEND
AWS permission gate (1–6)
Data / request
Conditional / return
Built from
Final query + answer
Decision
Claim Query Agent
Data store / prefix
Customer B boundary



Part	Key facts
KPI gate	One Claude Haiku call with structured output. If a preset fits, it fills the parameters (payer, payer class, month, CPT, CARC), which are checked against the data. Anything else goes to the agent.
Preset KPI queries	16 fixed SQL queries. They pass the SQL guard like any other SQL.
Claim Query Agent	The run worker itself (Fargate): Claude Haiku, Strands on Amazon Bedrock
Prompt	Cached: rules, EDI glossary, semantic schema with allowed values, tool and CLI reference, system prompt and history. Per question: the 3 most similar solved examples and the newest prompt.
First decision	Answer, ask a clarifying question, or escalate to the lossless tables with a reason, which is logged
Answer	Streams back with live steps. The server re-runs the final SQL, so the numbers come from the database.

05 · Chat backend



Part	Key facts
CloudFront + WAF	The only public web entry
Load balancer	Internal. Accepts traffic from CloudFront only.
Chat API (Fargate)	Gate 4 on every request. Queues one run per question and streams the answer over SSE. 3 unfinished runs per user.
Run queue (SQS)	One message per run
Run worker (Fargate)	The Claim Query Agent. Calls Bedrock and the DuckDB query service. 110 s run deadline.
Chat storage	Out of scope: answers stream live, and conversations are not stored

Network, operations and log retention are owned by IT.

Open decisions

Question	Default shown
SFTP: AWS Transfer Family or an EC2 host?	Transfer Family, writing straight into `raw/`
Does the audit record keep the SQL text?	Yes, the first 500 characters
Queue and run workers	SQS + Fargate run workers
Per-customer limits	Concurrency and token budgets still need values